RAK5814 WisBlock Crypto Module Datasheet

Overview

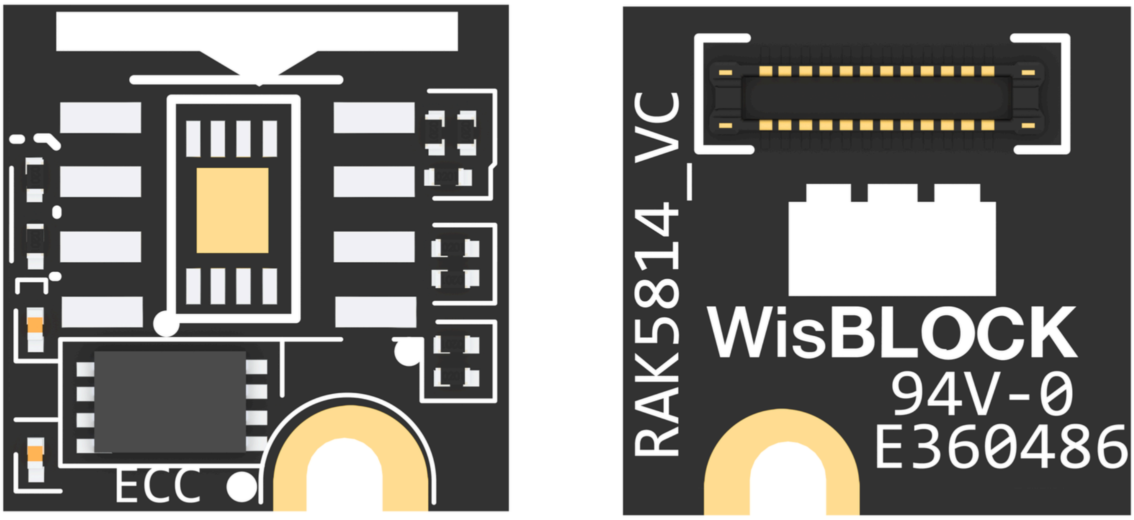

Figure 1: RAK5814 WisBlock Crypto Module

Description

RAK5814 is a secure element cryptography chip based on Microchip ATECC608A. It was designed to provide hardware-based key storage as well as encryption/decryption to various electronic products and devices. It is very low power and interfaced via I2C.

Features

- **Module Specifications**
  - Protected storage for up to 16 Keys, Certificates, or Data
  - Hardware support for Asymmetric Sign, Verify, and Key Agreement
  - Hardware support for Symmetric Algorithms
  - 3.3 V Power Supply

- I2C Interface (Max Speed: 100 kHz)
- Sleep Current less than 150 nA

- **Module Size**
    - 10 × 10 mm

Specifications

Mounting

Mounting to WisBlock Base

The RAK5814 crypto module can be mounted to the sensor slots A, B, C, and D of the WisBlock Base Board. **Figure 2** shows the mounting mechanism of the RAK5814 on a WisBlock base board module, such as the **RAK5005-O**.

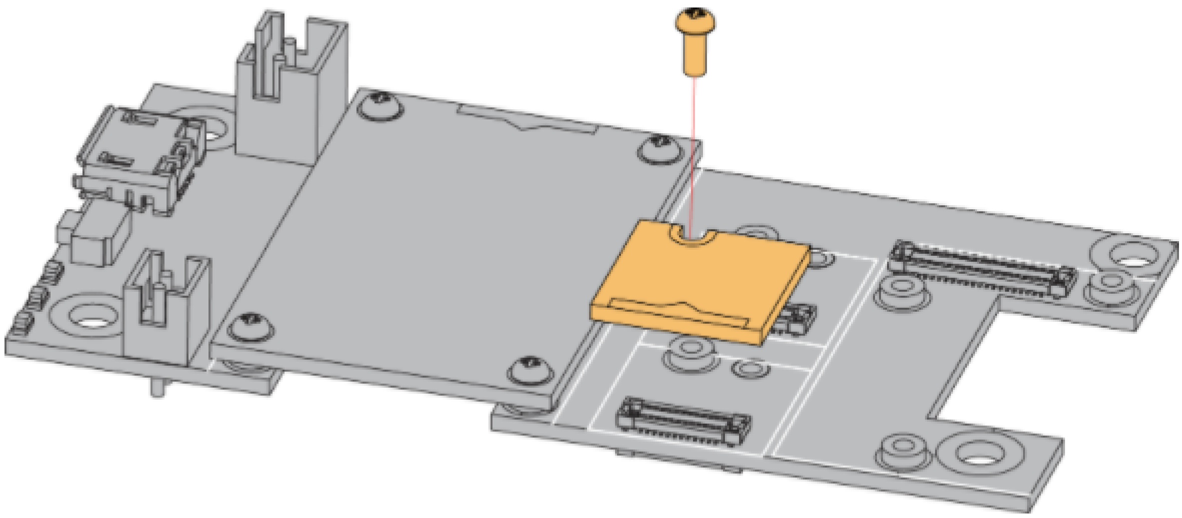

Figure 2: RAK5814 WisBlock Crypto Module Mounting

Hardware

The hardware specification is categorized into five (5) parts. It shows the chipset of the module and discusses the pinouts and their corresponding functions and diagrams. It also covers the electrical and mechanical parameters that include the tabular data of the functionalities and standard values of the RAK5814 WisBlock Crypto Module.

Chipset

Vendor	Part Number
Microchip	ATECC608A

Pin Definition

The RAK5814 WisBlock crypto module comprises a standard WisSensor connector. The WisSensor connector allows the RAK5814 module to be mounted to a WisBlock base board, such as RAK5005-O. The pin order of the connector and the pinout definition is shown in **Figure 3**.

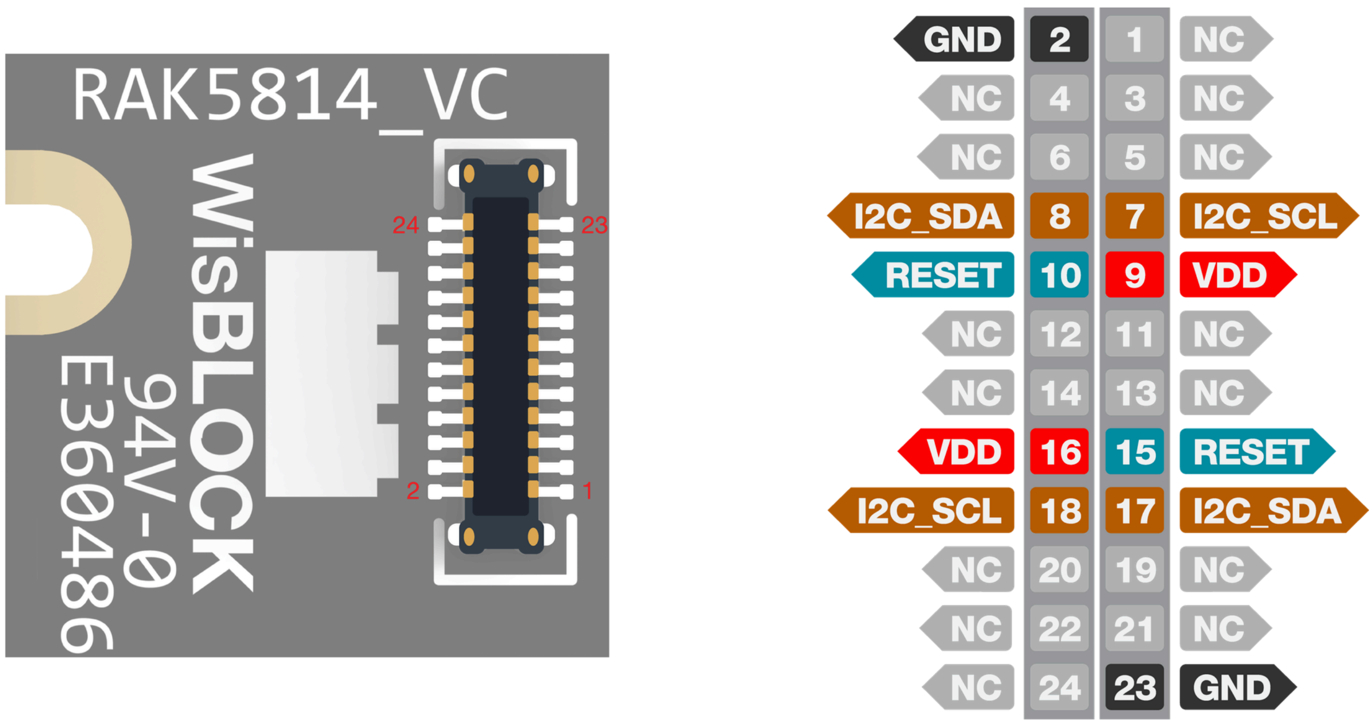

Figure 3: RAK5814 WisBlock Crypto Module Pinout

Electrical Characteristics

This table shows the RAK5814 module electrical characteristics.

Symbol	Description	Min.	Nom.	Max.	Unit
VDD	Power Supply Voltage	-	3.3	-	V

Symbol	Description	Min.	Nom.	Max.	Unit
Isleep	Sleep Current	-	-	150	nA
Icc	Active Power Supply Current	2	-	14	mA
Freq	I2C Speed	-	-	100	kHz

Mechanical Characteristic

Board Dimensions

Figure 4 shows the dimensions and the mechanical drawing of the RAK5814 module.

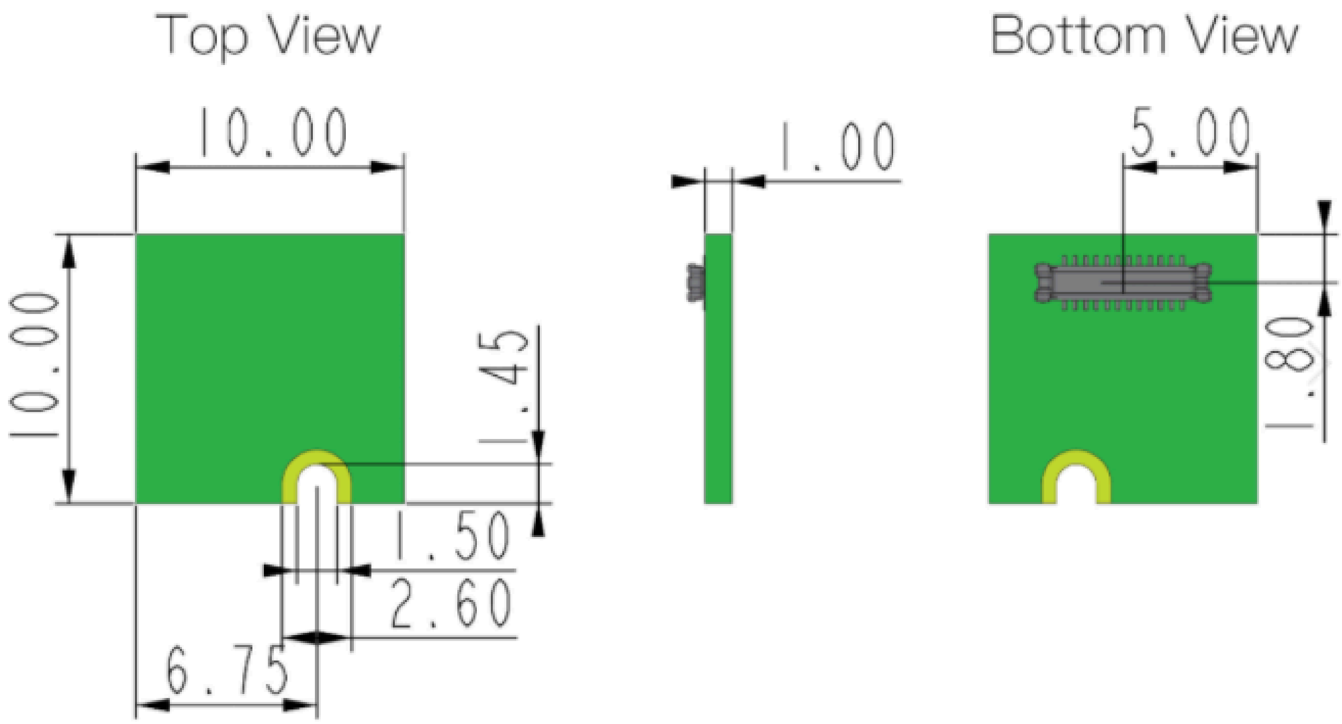

Figure 4: RAK5814 WisBlock Crypto Module Mechanical Drawing

WisConnector PCB Layout

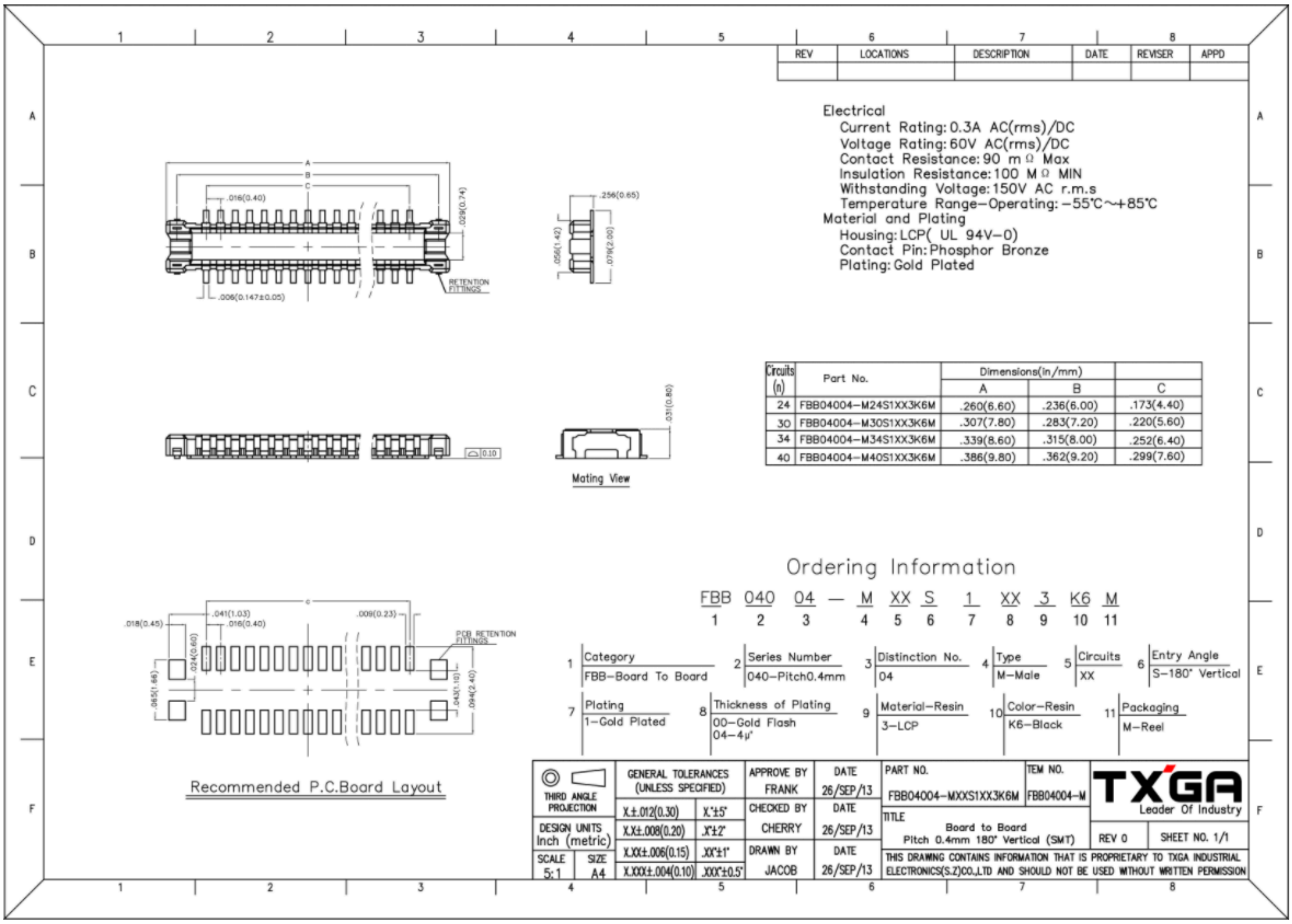

Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 6 shows the RAK5814 crypto module schematic. U1 (SOIC-8) and U2 (DFN8-2×3 mm) is reserved for different footprint crypto chip.

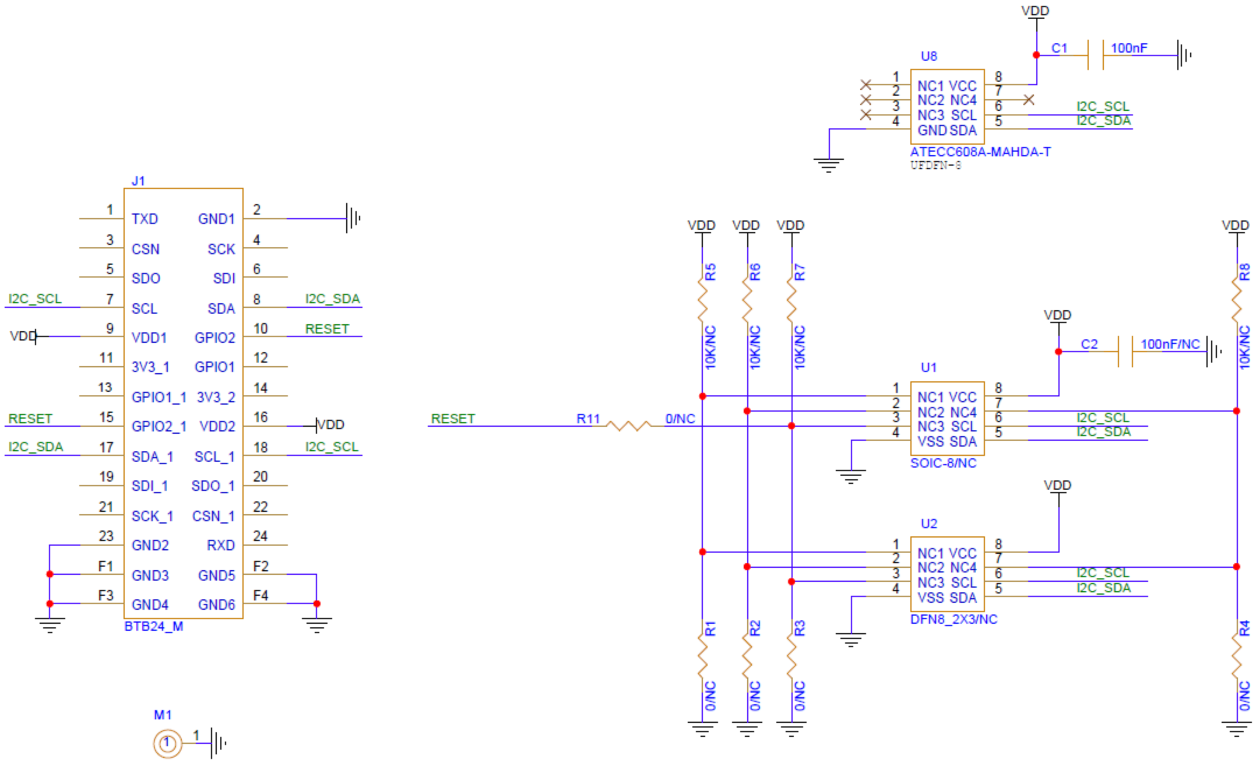

Figure 6: RAK5814 WisBlock Crypto Module Schematic